9 The use of satellites for communication using microwaves is relatively commonplace, for example in telephone links. The most useful sort of satellite orbit is a geostationary or geosynchronous one. It orbits the Earth which has a radius of 6.4×10^3 km. The microwave transmitting and receiving dishes do not have to constantly change their orientation, but can constantly point in the same direction.

One disadvantage, however, in the use of the geostationary satellite is that there will be regions on Earth where the satellite will appear to be below the horizon and cannot be seen at all. As shown in Fig. 9.1, there will be an angle, θ, above the equator, beyond which an observer on Earth cannot see the geostationary satellite.

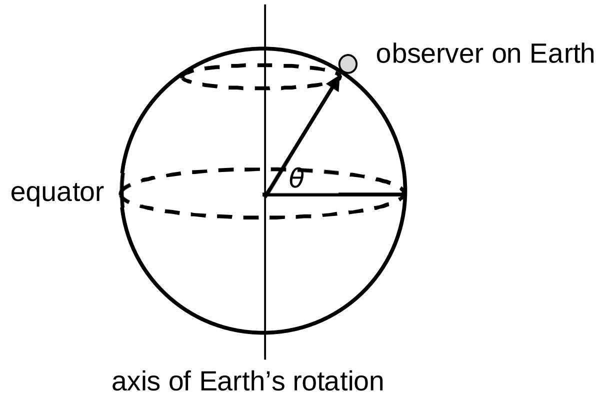


Fig. 9.1

Communication to and from moving ships and aircraft over long distances is obviously more difficult, even with geostationary satellites. Large ships move relatively slowly, and can carry a transmitting dish and the tracking equipment to keep it pointing in the right direction (assuming they know where they are on the Earth's surface accurately enough). However, small boats and aircraft have a weight and size problem; aircraft have the added complication of high speeds.

A type of microwave transmitter (and receiver) has been developed which can transmit in almost any carefully controlled direction without any moving parts, making it lighter and less vulnerable to mechanical damage than a conventional dish. The working principle is shown in Fig. 9.2.

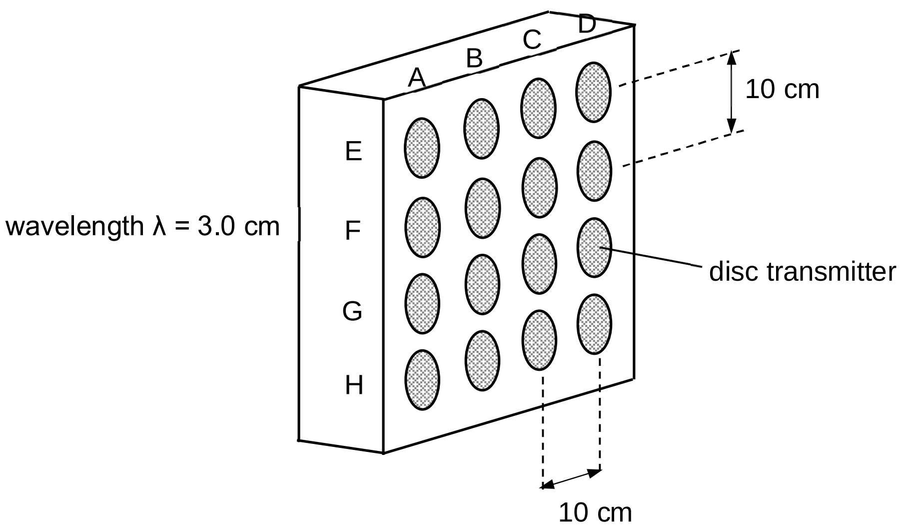


Fig. 9.2

For Examiner's Use

Fig. 9.2 shows a microwave transmitter which consists of 4 columns of discs, column A, B, C and D, and four rows of discs, row E, F, G and H. Each disc can have a different phase difference introduced into the signal before it is transmitted. For instance, if all the signals are in phase, there will be a strong signal sent directly in front of the array, but if row G is 1 cycle behind row H, row F 2 cycles behind row H and row E 3 cycles behind row H, a strong signal is sent upwards at an angle of about 17°. This is very similar to the way in which a diffraction grating works for light. Care has to be taken in choosing the spacing in the array, otherwise there could be very little signal in some directions.

(a) Calculate the time it takes a geostationary satellite to orbit the Earth once. [1]

time = 24.0 × 3600 = 8.64 × 10^4 s

(b) A geostationary satellite must be positioned directly above the Equator. Explain why this is so. [3]

This is to ensure that the satellite rotates about the same axis as the Earth. The line joining the object and the centre of Earth (which the gravitational force acts along) has to pass through the Equator and rests along the plane perpendicular to the Earth's axis of rotation.
If the object lies above other latitudes besides the equator, its axis of rotation will not coincide with that of the Earth's. As such it will be above different latitudes as it rotates and will not be geostationary.

(c) Using the data provided and your answer in **(a)**, show that the radius of a geostationary orbit is 4.2 × 10^4 km. Explain your working. [3]

The gravitational force on the satellite by Earth provides the centripetal force for it to orbit the Earth.
By Newton's 2nd Law of motion,

$$\Sigma F = ma$$

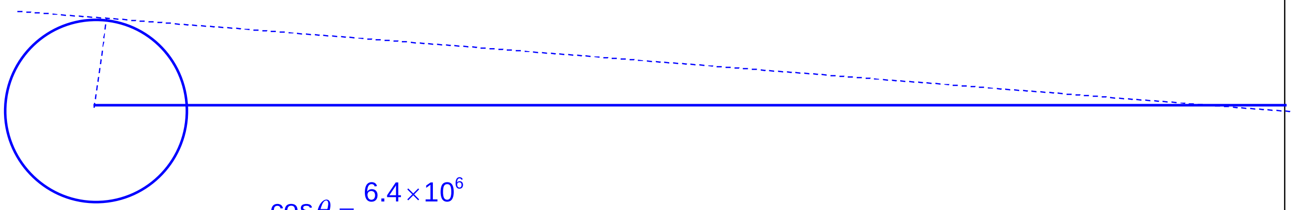
$$\frac{GM_E m}{r^2} = mr\omega^2$$

$$\frac{GM_E m}{r^2} = mr\left(\frac{4\pi^2}{T^2}\right)$$

$$r = \sqrt[3]{\frac{GM_E T^2}{4\pi^2}} = \sqrt[3]{\frac{gR_E{}^2 T^2}{4\pi^2}} = \sqrt[3]{\frac{(9.81)(6.4\times10^6)^2(86400)^2}{4\pi^2}} = 4.2\times10^7 \text{ m}$$

(Near the Earth's surface, $g = \frac{GM_E}{R_E{}^2} \Rightarrow GM_E = gR_E{}^2$)

(d) Estimate, by scale drawing or otherwise, the angle θ above the Equator at which a receiving dish can be and still "see" the satellite. [3]



$$\cos\theta = \frac{6.4\times10^6}{4.2\times10^7}$$

$$\theta = 81°$$

accept 75° to 85° for scale drawing; 81° for exact calculation

(e) Show that the angle of 17° given in the passage is correct. [1]

$$d\sin\theta = n\lambda$$

$$\theta = \sin^{-1}\left(\frac{n\lambda}{d}\right)$$

$$= \sin^{-1}\left(\frac{1\times 0.030}{0.10}\right)$$

$$= 17°$$

(f) A strong signal can be sent out if the phase relationships were as follows; vertical column A all in phase, column B all 1/3 of a cycle later, column C all 2/3 of a cycle later and column D all 1 cycle later.

(i) Calculate the path difference of the waves between adjacent column. [1]

Path difference = 1/3 × 0.03. = 0.010 m

(ii) Hence, determine the angle and direction, of this strong signal. [2]

$$d\sin\theta = n\lambda$$

$$\theta = \sin^{-1}\left(\frac{n\lambda}{d}\right)$$

$$= \sin^{-1}\left(\frac{\frac{1}{3}\times 0.030}{0.10}\right)$$

$$= 5.7°$$

Direction to towards column D.

(g) Suggest, in principle, and without technical details, how a ship's communication with satellites could be used as a navigation aid, so that it can know accurately where its position is. [3]

The ship can transmit signals in all directions by varying the phase difference between the columns as well as the rows of transmitters. When the signal is received by a satellite in a known position, the satellite can transmit a signal back to notify the ship. The ship can then use its own orientation on Earth (using a compass) and the phase differences between the columns and rows to calculate the exact location it is on the Earth's surface.

(h) "Care has to be taken in choosing the spacing in the array". In this "phased array" setup, it is essential that each disc is only a few wavelengths apart from its neighbouring discs. Suggest why the discs cannot be positioned too close together or too far apart. [2]

If the discs are positioned too close together (much less than 1 wavelength), signals can only be detected at certain angles.
If the discs are positioned too far apart, a small phase difference between the discs will not change the angle of the signal significantly.

End of Paper